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NA-A7 solution

1. (0,0) (1,3) (2,3) (5,6)

(a) $y = ax+b$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 5 & 1 \end{pmatrix} \quad Y = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 6 \end{pmatrix} \quad Z = \begin{pmatrix} a \\ b \end{pmatrix}.$$

$$\min (XZ-Y)^T(XZ-Y)$$

$$\Rightarrow Z = (X^TX)^{-1}X^TY$$

$$= \left[\begin{pmatrix} 30 & 8 \\ 8 & 4 \end{pmatrix}\right]^{-1} \begin{pmatrix} 39 \\ 12 \end{pmatrix}$$

$$= \frac{1}{56}\begin{pmatrix} 4 & -8 \\ -8 & 30 \end{pmatrix}\begin{pmatrix} 39 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{15}{14} \\ \frac{6}{7} \end{pmatrix} \Rightarrow y = \frac{15}{14}x + \frac{6}{7}$$

~~(b) $y = ax^2+bx+c$~~

~~$\min (c-0)^2 + (a+b+c-3)^2 + (4a+2b+c-3)^2 + \cdots$~~

~~let $X = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \\ 25 & 5 & 1 \end{pmatrix}$ $Y = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 6 \end{pmatrix}$ $Z = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$.~~

~~$Z = (X^TX)^{-1}X^TY =$~~

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(b) $y = ax^2 + bx + c$

$$\min f(a,b,c) = (c-0)^2 + (a+b+c-3)^2 + (4a+2b+c-3)^2 + (25a+5b+c-6)^2$$

$$\frac{\partial f}{\partial a} = 2(a+b+c-3) + 8(4a+2b+c-3) + 50(25a+5b+c-6)$$

$$\frac{\partial f}{\partial b} = 2(a+b+c-3) + 4(4a+2b+c-3) + 10(25a+5b+c-6)$$

$$\frac{\partial f}{\partial c} = 2c + 2(a+b+c-3) + 2(4a+2b+c-3) + 2(25a+5b+c-6)$$

$$\Rightarrow \frac{\partial f}{\partial a} = \frac{\partial f}{\partial b} = \frac{\partial f}{\partial c} = 0$$

$$\Rightarrow \begin{cases} 1284a + 686b + 60c - 330 = 0 \\ 268a + 60b + 16c - 78 = 0 \\ 60a + 16b + 8c - 24 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a \approx -0.1657 \\ b \approx 1.9475 \\ c \approx 0.3481 \end{cases} \Rightarrow y = -0.1657x^2 + 1.9475x + 0.3481$$

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(C) $y = ax^3 + bx^2 + cx + d$

~~f(a,b,c,d)~~

$$X = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \\ 125 & 25 & 5 & 1 \end{pmatrix} \quad Y = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 6 \end{pmatrix} \quad Z = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$\Rightarrow Z = (X^TX)^{-1}X^TY \Rightarrow a \approx 0.35 \quad b \approx -2.55 \quad c \approx 5.20$

$d = 0.$

$\Rightarrow y = 0.35x^3 + (-2.55)x^2 + 5.2x$

Lagrange interpolating:

$$y = 3 \cdot \frac{(x-0)(x-2)(x-5)}{(1-0)(1-2)(1-5)} + 3 \cdot \frac{(x-0)(x-1)(x-5)}{(2-0)(2-1)(2-5)} + 6 \cdot \frac{(x-0)(x-1)(x-2)}{(5-0)(5-1)(5-2)}$$

$$= \frac{7}{20}x^3 - \frac{51}{20}x^2 + \frac{26}{5}x$$

the polynomials are same.

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2. (a)

0	0		
$\frac{\pi}{2}$	1	$\frac{1-0}{\frac{\pi}{2}-0}=\frac{2}{\pi}$	
π	0	$\frac{0-1}{\pi-\frac{\pi}{2}}=-\frac{2}{\pi}$	$\frac{-\frac{2}{\pi}-\frac{2}{\pi}}{\pi-0}=-\frac{4}{\pi^2}$

$$\therefore P_2[x]=0+\frac{2}{\pi}(x-0)+(-\frac{4}{\pi^2})(x-0)(x-\frac{\pi}{2})$$

(b) $P_2[\frac{\pi}{4}]=\frac{3}{4}$

(c) interpolation error:

$$f(x)-P(x)=\frac{(x-x_1)(x-x_2)(x-x_3)}{3!}f^{(3)}(c)$$

so

$$|\sin x-P_2[x]|\le\left|\frac{x(x-\frac{\pi}{2})(x-\pi)}{3!}\right|\cdot|1|$$

$$\Rightarrow\left|\sin\frac{\pi}{4}-P_2[\frac{\pi}{4}]\right|\le\frac{\pi^3}{128}\approx 0.2422$$

(d) $\left|\frac{3}{4}-\sin\frac{\pi}{4}\right|=\left|\frac{3}{4}-\frac{\sqrt{2}}{2}\right|\approx 0.04289$.

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3. x_j $(j=0,1,2,\cdots n)$

(a) Show that $\sum_{j=0}^{n} x_j^k L_j(x) \equiv x^k$ $(k=0,1,2,\cdots,n)$

we know

$$L_j(x_i) = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$$

let $f(x) = \sum_{j=0}^{n} x_j^k L_j(x) - x^k$

so $f(x_j) = 0 \quad j=0,1,2,\cdots,n$

$\Rightarrow$ $n+1$ distinct roots.

when $k=0,1,2,\cdots,n$, $\partial(f(x)) \le n$ contradiction!

$\Rightarrow f(x) \equiv 0$

$\Rightarrow \sum_{j=0}^{n} x_j^k L_j(x) \equiv x^k$

(b) Show that $\sum_{j=0}^{n} (x_j - x)^k L_j(x) \equiv 0 \quad k=0,1,2,\cdots,n.$

let

$$g(x) = \sum_{j=0}^{n} (x_j - x)^k L_j(x) = \sum_{j=0}^{n} x_j^k L_j(x) + -C_k^1 x \sum_{j=0}^{n} x_j^{k-1} L_j(x) +$$

~~If $g(x_j) = 0$~~

$$\cdots + (-1)^{k-1} x^{k-1} \sum_{j=0}^{n} x_j L_j(x)$$

$$+ (-1)^k x^k \sum_{j=0}^{n} L_j(x)$$

consider $x^m \sum_{j=0}^{n} x_j^{k-m} L_j(x) \quad m=0,1,\cdots,k$

$$\equiv x^m \cdot x^{k-m} = x^k \text{ (from (a))}$$

~~m roots of 0.~~

so $g(x) = x^k - C_k^1 \cdot x^k + C_k^2 \cdot x^k - \cdots + (-1)^k x^k$

$$= x^k (1 - C_k^1 + C_k^2 - \cdots + (-1)^k)$$

$$= x^k (1 + (-1))^k = 0.$$

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